

Truth Learning in a Social Network Under Random Decision Orderings

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Background

- In social settings, we often make decisions based on what we see others do, combined with our own private beliefs
- This can lead to an information cascade, in which people blindly follow the pack without contributing their private information:

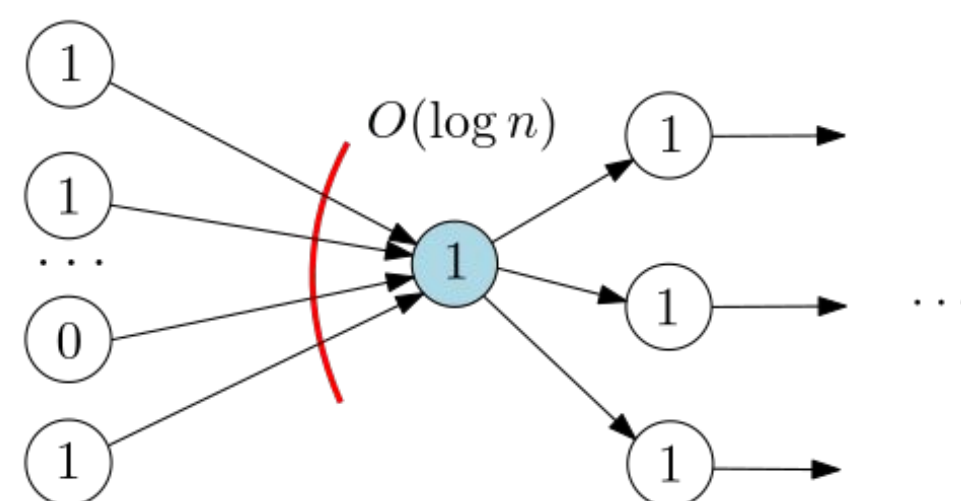


Central Question

What are the necessary/sufficient conditions for a social network to support asymptotic truth learning under random orderings?

Setting

- Networks are infinite families $\mathcal{F} = \{G_n, \sigma_n\}_n$ of graphs G_n and orderings σ_n . Agents (vertices) can only see their neighbor's decisions
- Ground truth $\Theta \in \{0, 1\}$ is an unknown random variable
- Agents are given independent private signals $p_v \in \{0, 1\}$ each correlated with Θ with known parameter $q > 1/2$
- Each agent broadcasts their prediction in an order according to σ
- Agents are perfectly Bayesian & predict the most probable truth given their knowledge of G , σ , q , and their neighbors' signals
- The learning rate $\ell(v)$ of a vertex under an ordering σ is the probability that it correctly predicts the ground truth. The learning rate $L(G, \sigma)$ is the average learning rate of all vertices.



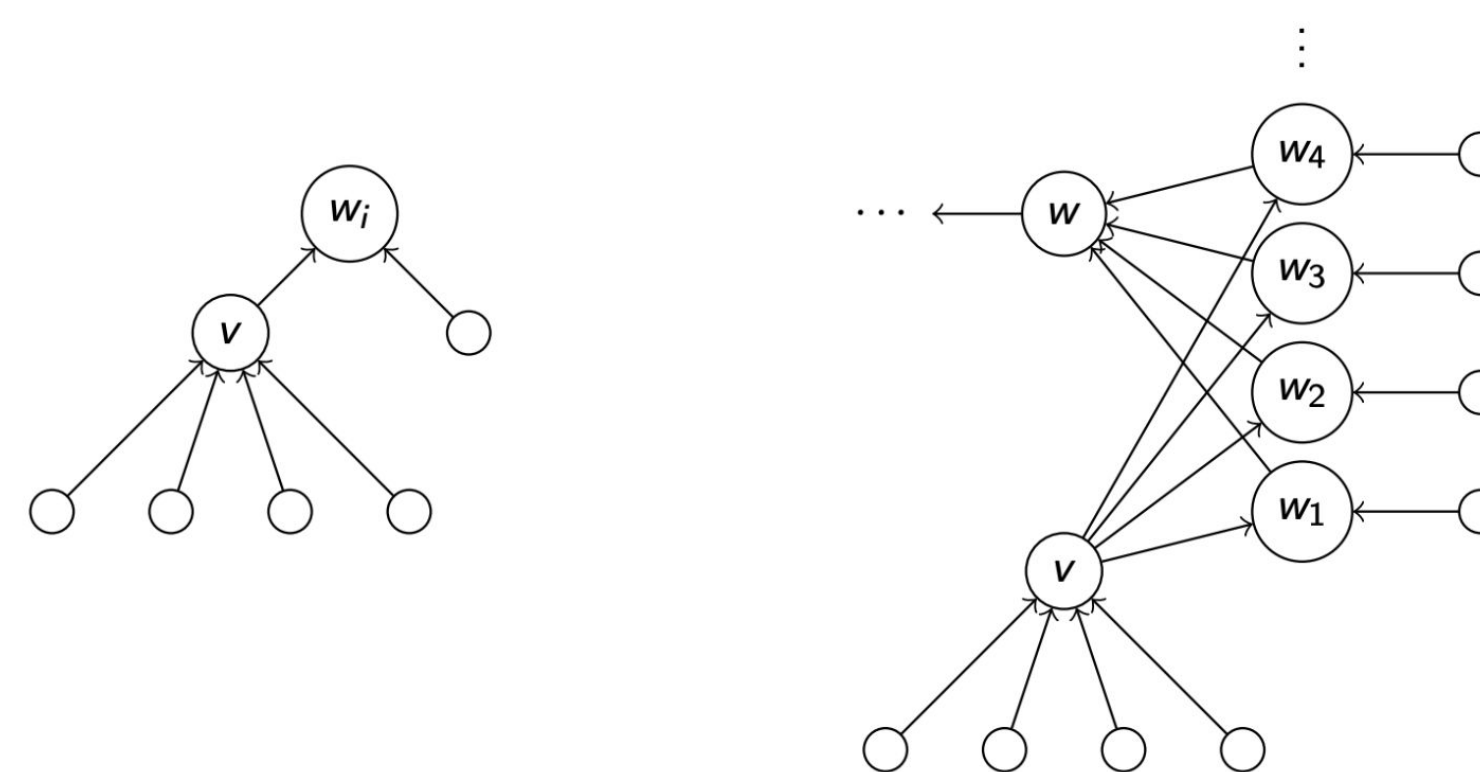
Asymptotic truth learning: $L(G_n, \sigma_n)$ approaches 1 as $n \rightarrow \infty$.

- The random learning rate $L(G)$ of a graph G with n vertices is the average learning rate over all $n!$ orderings.
- A network achieves *random* learning if $L(G_n) \rightarrow 1$ as $n \rightarrow \infty$.

Motivating Example: Fragile Social Networks

Example

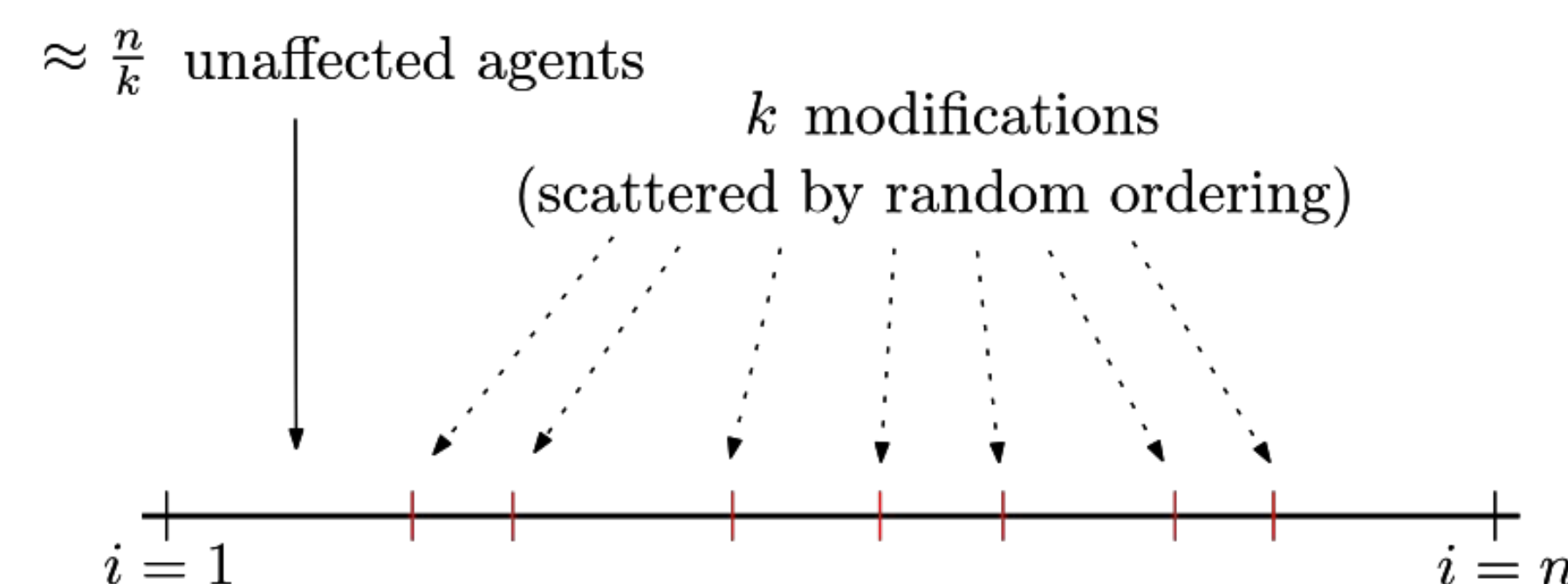
There exists a network $\mathcal{F}$ that learns iff each agent's private signal strength is $q < q_0$, for some threshold $q_0 \approx 0.79$.



Robustness of Random Order Learning

Claim (monotonicity)

Let $\ell(v|i)$ be the random-order learning rate of v conditioned on being at index i . Then for any $i \in [n-1]$, $\ell(v|i) \leq \ell(v|i+1)$.

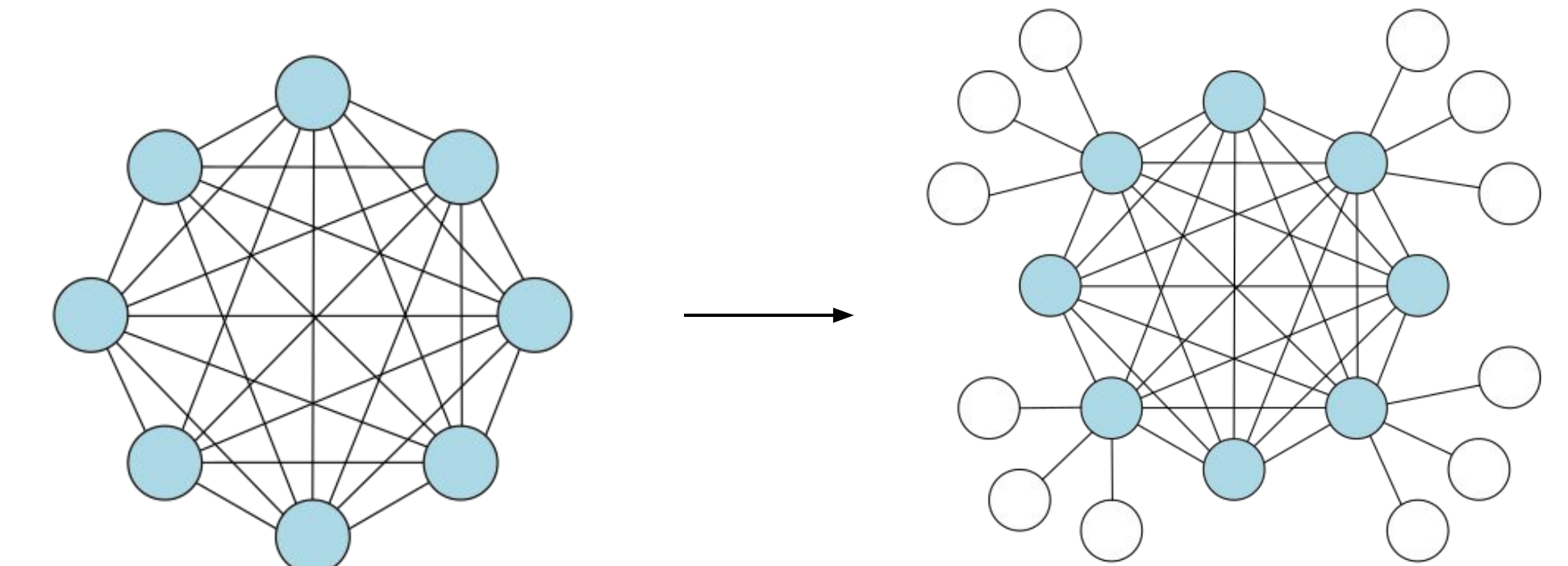


Theorem

Given a graph G with random-order learning rate $1 - \varepsilon$, after k modifications, G obtains learning rate $1 - (k+1)\varepsilon$

Engineering Random Order Learning Networks

Idea: choose specific vertices to 'boost' by connecting them to $\omega(1)$ independent guinea pigs.



Definition

Let $f(v, t) : V \times (0, 1) \rightarrow \{0, 1\}$ be a "learning oracle" that outputs 1 if v learns at rate at least t under random orderings, and 0 otherwise.

Theorem

Given efficient query access to f , there exists an algorithm running in polynomial time that modifies a graph to obtain random-order learning. Furthermore, the number of modifications needed is a $g(n) \log n$ -approximation of the optimum, for any function $g(n) = \omega(1)$.

Future Work

Open Problem

Is determining whether a graph/vertex obtains a learning rate above a certain threshold NP-hard/APX-hard?

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References

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